Brandon Yifan Yang

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Education

- **University of Pennsylvania** Philadelphia, PA
  M.S.E. in Robotics; GPA: 3.7/4.0 *May 2027*
- **University of Virginia** Charlottesville, VA
  B.S. in Computer Science; Highest Distinction; GPA: 3.9/4.0 *May 2025*
- **Selected Coursework**: Learning in Robotics, Reinforcement Learning, Probabilistic Machine Learning, GPU Programming & Architecture, Physical Intelligence, Natural Language Processing

Experience

Research Lead San Francisco, CA
AfterQuery (YC W25) *May 2026 - Present*

- Led development of custom UMI hardware and a distributed robot data collection system, integrating Rust-based device-side capture and LiDAR tracking with a server-side SLAM pipeline for end-effector tracking and automated storage with Hugging Face dataset management.
- Led a 6-person research team building a full-stack AI-agent search benchmark platform; collected tasks from external experts, built custom quality-control workflows, and automated evaluations across frontier agent harnesses and models.

Embodied Intelligence Research Intern Beijing, China
Spirit AI *May 2025 - Aug 2025*

- Trained and evaluated VLA policy variants in PyTorch across simulation and real dual-arm robot experiments, comparing contrastive objectives, observation-noise curricula, reduced guidance inputs, and mixture-of-experts architectures.
- Designed and implemented a DAgger data-collection pipeline for VLA models, including operator-takeover logic and corrective-demonstration workflows used for post-training and evaluation.
- Built a reproducible Docker/Slurm pipeline for VLA simulation, training, and evaluation workflows; open-sourced code and datasets on GitHub.

Research Experience

Perception, Action, & Learning Lab, University of Pennsylvania Philadelphia, PA
Advisor: Dinesh Jayaraman *Aug 2025 - Present*

- Implemented demonstration retrieval for $\pi_{0.5}$ VLA policies, integrating robot data retrieval into training and test-time inference for tabletop manipulation experiments. (*CoRL 2026*)
- Engineered data transformation, training, and evaluation pipelines for retrieval-based VLA variants using JAX, PyTorch, Hugging Face, Docker, Slurm, and Ray.
- Designed hidden-activation collection and steering infrastructure for VLA analysis across MetaWorld, LIBERO, RoboCasa, and real Franka hardware experiments. (*ICML 2026 Workshop; NeurIPS 2026*)

Learning and Interactive Robotics, University of Virginia Charlottesville, VA
Advisor: Yen-Ling Kuo *Aug 2024 - May 2025*

- **SkillVLA**: Developed a skill-conditioned VLA model for language-conditioned manipulation, using subgoal instructions and a learned skill library to structure action generation.
- Presented SkillVLA as an oral talk at the 2024 UVA LLM Workshop (slides), earning the Audience Choice Award (top 3 of 28 presentations).
- **Contrastive Learning for Robot Manipulation**: Trained contrastive objectives over action sequences to learn behavior-grounded visual embeddings for visuomotor policies under varied camera poses and object appearances. (CoRL 2025)
- Modified simulation environments and trained PyTorch visuomotor policies to evaluate learned behavior-grounded embeddings.

Collaborative Robotics Lab, University of Virginia Charlottesville, VA
Advisor: Tariq Iqbal *May 2022 - May 2024*

- **Grounded Location for Object Manipulation (GLOMA)** [code]: Led a team of 3 in developing a language-grounded image-editing pipeline for object relocation goals in robotic manipulation settings.
- Integrated language grounding with bounding-box-guided visual editing for object relocation.
- Collected and annotated a dataset for fine-tuning pre-trained language and vision models.
- **Centralized Multi-Agent RL for Collaborative Tasks** [code]: Implemented offline and online centralized MARL training for long-horizon robotic bolt-screwing tasks in Isaac Gym.
- Designed custom multi-agent reward functions for task completion and collaboration, improving simulated task success by 20% over the project baseline.

Publications

Jiani Huang*, **Brandon Y. Yang***, Amish Sethi, Yuchen Zheng, Christopher Watson, Jianing Qian, Mayur Naik, and Dinesh Jayaraman. "Retrieval-Augmented Vision-Language-Action Model." Under review at *CoRL 2026*.

Miranda Muqing Miao*, Subin Kim*, **Brandon Y. Yang***, Dinesh Jayaraman, and Lyle Ungar. "Contrastive Conceptor Activation Steering (COAST): Unlocking Vision-Language-Action Models through Hidden States." Accepted poster, *ICML 2026 Mechanistic Interpretability Workshop*; submitted to *NeurIPS 2026*. [Workshop][arXiv]

Amish Sethi*, **Brandon Y. Yang***, Michael Evans, Spencer Mateega, and Sam Jung. "Position: Trust No Multi-Turn Benchmark Without Measuring Its User Simulator." Submitted to *NeurIPS 2026 Position Paper Track*.

Sung-Wook Lee, Xuhui Kang, **Brandon Y. Yang**, and Yen-Ling Kuo. "CLASS: Contrastive Learning via Action Sequence Supervision for Robot Manipulation." *Conference on Robot Learning*, pp. 4743–4766. PMLR, 2025. [Website][arXiv][PDF]

*Equal contribution among starred authors.

Software Projects

openpi-eval [GitHub, Dataset]: Built OpenPI evaluation infrastructure for a NeurIPS submission, supporting reproducible `pi05` rollouts across MetaWorld, LIBERO, LIBERO-Pro, RoboCasa, and RoboLab, plus `pi0_fast` support where released and server-side inference batching.

openpi-cuda [GitHub]: Developed custom CUDA kernels with C++/Python bindings for $\pi_{0.5}$ VLA inference, reducing latency by 18.5ms versus the PyTorch baseline.

notie-markdown [Website]: Built a React/TypeScript markdown renderer with equation previewing, graphing, and code execution; shipped with a Python/Flask backend on Vercel and Heroku.

SmartOH [GitHub]: Built an office-hours queue management system with real-time queue updates, notifications, and analytics using Next.js, TypeScript, Python, and FastAPI; deployed with Vercel, AWS EC2, and CI/CD.

Skills

Programming: Python, C++, CUDA C++, TypeScript/JavaScript, Java

ML & Robotics: PyTorch, JAX, Hugging Face, CUDA, Ray, Slurm, robot learning, VLA models

Web & Infrastructure: React, FastAPI, Flask, Docker, AWS EC2, Vercel, `uv`, `ruff`

Last Updated: July 19, 2026